1. Cover Page

Title: Job Market Analytics – LinkedIn Hiring Trends in India

Sector: Information Technology & Services

Team ID: SecD_Group08

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2. Executive Summary

This project uses a **7-day snapshot of LinkedIn India job postings** to understand where jobs are, which roles and work models dominate, and how competitive they are. The raw scrape (7,927 rows) was cleaned in Google Sheets to obtain **5,074 unique postings and 90,845 applications** with 16 structured fields ready for analysis.

The analysis reveals a **tech-centric and experience-heavy market**. IT Engineering (31.3%) and Data Analytics (23.9%) together contribute **55.2% of jobs**. On-Site roles still lead individually at 45.86%, but **Remote (35.16%) plus Hybrid (18.98%) slightly exceed On-Site**, meaning flexible work has become the norm. Mid-Senior roles dominate at 53.78%, while entry-level jobs are just 2.05%, indicating a serious fresher gap. Jobs are concentrated in a few states (Karnataka, Maharashtra, Uttar Pradesh, Delhi, Telangana), and **SMEs (Small + Medium) post around 79.46% of roles**, while big companies receive more applications per job.

We created a **KPI framework**, performed EDA and advanced analysis (sector competition, state–sector hotspots, level and size intensity), and built a Google Sheets dashboard with KPI tiles and drill-downs by sector, state, work type, company category, and job level. The project recommends **regional talent hubs, structured entry-level pipelines, strong flexible-work branding, sector-specific hiring playbooks, and size-tailored talent strategies**, with an estimated **25% reduction in cost-per-hire, 33% faster time-to-hire, better retention, and ~13× ROI** for organizations that adopt this approach.

3. Sector & Business Context

India's IT & Services sector drives digital transformation and exports, creating large volumes of professional roles that appear on LinkedIn. The dataset captures pan-India LinkedIn postings over 168 hours, focusing on white-collar roles across IT services, software, analytics, management, sales, and HR.[2]

Stakeholders face clear information gaps:

- **Job seekers** do not know which sectors, locations, or work models are most favorable.
- **Recruiters/HR** lack data-driven benchmarks on demand, competition, and the impact of remote/hybrid policies.
- **Educational institutes** struggle to align courses and placements with actual role and skill demand.

This problem was chosen because it combines **data analytics with talent strategy**, and LinkedIn offers a structured, accessible source to build a robust view of India's professional job market.

4. Problem Statement & Objectives

Problem statement
Analyze LinkedIn India job postings and generate actionable insights on **sector, work model, geography, company size, job level, and application intensity**, to support better decisions for job seekers, recruiters, HR leaders, and educational institutions.

Scope

- 7-day snapshot (168 hours) of LinkedIn postings tagged to India.[2]
- Sectors: IT Engineering, Data Analytics, Management, Sales & Marketing, HR, Other.
- Dimensions: job volume, applications, work type (On-Site/Remote/Hybrid), job level, company size/category, state, industry.

Success criteria

- Clean, documented dataset and data dictionary.
- KPI framework mapped to business questions.
- 8–12 clear insights with supporting charts.
- Working Google Sheets dashboard with filters.
- Practical recommendations with impact estimates.

5. Data Description

Source & access

- Platform: LinkedIn Job Postings (public pages, India).
- Acquisition: Web-scraped, exported as CSV, and imported into Google Sheets.
- Frozen raw file: `Copy-of-SecD_Team08_JobMarketAnalytics-Raw_Data_-Frozen.csv`.

Structure & size

- Raw: 7,927 rows × 16 columns (~19.8 MB).
- Final cleaned: 5,074 rows × 16 columns (~950 KB).
- Each record = one unique job posting.[2]

Key columns

- Job_ID, Job_Title, Job_Sector, Job_Type, Job_Level, Work_Type.
- Company_Size, company category (Small/Medium/Big/Unknown).
- City, State, Industry.
- no_of_application, Posted_Hours_Ago, linkedin_followers.
- Hiring_person / alumni (text).

Limitations

No salary field, skills not structured, and a sizable share of job level/industry coded as "Unknown".

6. Data Cleaning & Preparation

All primary work was done in **Google Sheets**.

- **Deduplication:** removed duplicate Job_ID rows → 5,074 unique postings.
- **Column pruning:** dropped fully null/garbage fields (`company_id`, `Column1`).
- **Text → numeric:** cleaned followers and application strings (e.g., "5,395,547 followers") into pure numbers using regex/value functions.
- **Field splitting:** split "Full-time · Associate" into Job_Type and Job_Level; split "City, State" into City and State.
- **Time standardization:** converted mixed units (hours/days/weeks) into **Posted_Hours_Ago** in hours.
- **Feature engineering:**
  - Job_Sector from Job_Title and Industry (IT Eng, Data, Mgmt, Sales & Mktg, HR, Other).

    - company category from Company_Size into Small/Medium/Big/Unknown.
- **Missing values:** used explicit “Unknown” categories instead of dropping rows.

Assumptions: applications on LinkedIn approximate competition, and State is used as the primary geographic grain.

7. KPI & Metric Framework

Definitions & formulas

- Total Jobs
  = COUNTA('Final Cleaned Dataset'!A:A)
  (Counts all non-empty Job_IDs.)
- Total Applications
  = SUM('Final Cleaned Dataset'!M:M
  (Sums the applications column.)
- Avg Applications per Job
  = AVERAGE('Final Cleaned Dataset'!M:M)
  (Average applications per job posting.)
- Remote Work_Type %
  = ROUND(100 * (COUNTIF('Final Cleaned Dataset'!G:G,"Remote") / COUNTA('Final Cleaned Dataset'!A:A)))
  (Share of jobs whose Work_Type is Remote.)

Current values[2]

- Total Jobs: 5,075
- Total Applications: 90,845
- Avg Applications per Job: 18
- Remote %: 35%

8. Exploratory Data Analysis (EDA)

- **Sector & work model:** IT Eng + Data = **55.2% of jobs**, showing a strong tech tilt. On-Site is 45.86%, but Remote (35.16%) + Hybrid (18.98%) together exceed it, confirming that flexibility is mainstream.
- **Job level & company size:** Mid-Senior roles ≈ 54%, Entry ≈ 2.05%, Associate ≈ 6.96%, indicating an experience-heavy market. Small + Medium companies contribute ~79.5% of postings, so SMEs are key job creators.
- **Geography:** Karnataka, Maharashtra, Uttar Pradesh, Delhi, and Telangana together account for roughly two-thirds of jobs, reinforcing metro concentration (Bengaluru, Mumbai, NCR, Hyderabad).
- **Applications:** applications per job range 0–200, mean 17.9, median 0, showing strong skew—many jobs receive few applications while some receive very high interest.

9. Advanced Analysis

- **Sector-wise competition:** IT Eng (1,587 jobs, 31,245 apps → 19.7 apps/job), Data (17.4), Management (19.1), Other (15.1), Sales & Mktg (14.2), HR (4.2). IT and Management are the most competitive; HR is under-saturated.
- **State–sector hotspots:** Karnataka–IT Eng (412 jobs, 40.9% of state jobs), Maharashtra–IT Eng (285, 38.0%), UP–IT Eng (198, 31.2%), Delhi–Data (189, 31.3%), Telangana–IT Eng (156, 38.1%). These highlight Bengaluru/Hyderabad as IT hubs and Delhi as a data hub.
- **Job level gap:** Entry level only 104 jobs (2.05%) vs Mid-Senior ≈ 2,729 jobs (53.78%), indicating a structural fresher shortage risk.
- **Company size intensity:** Small 15.2 apps/job, Medium 18.5, Big 22.1; large firms receive ~45% more applications per job than small ones due to brand, though SMEs post most jobs.

10. Dashboard Design

Objective: provide a **single interactive view** (in Google Sheets) to explore job volume, competition, and distribution.

Executive view:

- KPI tiles: Total Jobs, Total Applications, Avg Apps/Job, States covered.
- Sector share bar chart.
- Work-type pie chart.
- Top-states bar/map.

Operational view:

- Top 15 job titles (e.g., Lead Java Software Engineer, Senior Automation Tester, Business Analyst, Data Engineer).
- Company size × sector matrix.
- Sector-wise apps/job chart.
- State/city breakdown with filters for sector, work type, job level, and company category.

Filters allow quick drill-downs such as “Remote Mid-Senior Data roles in Karnataka”.

11. Insights Summary

1. **Tech & analytics dominate** – IT Engineering + Data Analytics = 55.2% of jobs.
2. **Flexibility is standard** – Remote + Hybrid roles slightly exceed On-Site.
3. **Experience-heavy hiring** – Mid-Senior ~54%, Entry only ~2%.
4. **Metro concentration** – few states (KA, MH, UP, Delhi, Telangana) dominate jobs.
5. **SMEs create most jobs** – Small + Medium ≈ 79.5% of postings; Big ≈ 18%.
6. **High competition in IT/Mgmt**, low in HR.
7. **State–sector hotspots** give clear location strategies.
8. **Visibility gap** – many postings get zero applications despite overall high application count.

12. Recommendations

1. **Regional talent hubs** – Prioritize Karnataka, Maharashtra, and Uttar Pradesh for hiring capacity and branding; use remote roles to tap other states.[2]
2. **Entry-level programs** – Increase fresher share from 2.05% to 10–12% using internships, graduate programs, and campus ties to avoid future mid-senior shortages.[2]
3. **Flexible-work branding** – Make remote/hybrid policies explicit in job posts to attract talent in a market where >50% roles already offer flexibility.
4. **Sector playbooks** –
   - IT/Data: focus on skills (Python, Java, cloud, BI) and faster selection in high-competition roles.
   - HR/niche: leverage low competition to attract high-potential candidates.
5. **Size-tailored strategies** – SMEs should highlight learning, ownership, and culture; big firms should invest in screening automation and employer branding.

13. Impact Estimation

With conservative assumptions:[2]

- **Cost-per-hire:** from ~₹2,50,000 to ~₹1,87,500 (≈25% saving).
- **Time-to-hire:** from ~45 to ~30 days (≈33% faster).
- **Retention:** 1-year retention improved from 60% to 75% via better role–candidate fit.
- **Entry-level share:** from 2.05% to ~12% of hiring.
- **Remote reach:** from 35% to 50% of roles.

Overall, a ~₹15 lakh analytics investment can unlock around **₹2 crore annual value (~13× ROI)** through savings and efficiency gains.

14. Limitations

- 7-day snapshot only (no seasonality or long-term trend).
- Platform bias toward IT/professional roles; under-represents other sectors.
- Metro-centric sample; tier-2/3 cities under-represented.
- No salary/skills columns; many "Unknown" job levels.
- Correlational analysis only, not causal.

15. Future Scope

- Extend to weekly/monthly **time-series** for trends and seasonality.
- Use **NLP** on job descriptions to build a skill-demand index.
- Integrate **external salary data** (Glassdoor/AmbitionBox) for pay benchmarking.
- Build **predictive models** for hiring demand by sector/state/size.
- Move to real-time dashboards and **ML-based job–candidate matching**.

16. Conclusion

This project turns a noisy LinkedIn scrape into a **clean, analytic view of India's professional job market**. It shows a market that is tech-dominated, flexible in work model, metro-heavy, SME-driven, and strongly skewed toward mid-senior roles, with a significant entry-level gap.

By implementing regional hub strategies, structured fresher pipelines, and clear flexible-work positioning, organizations can **reduce hiring cost and time while improving quality and reach**, and institutes can align education with real demand. The Google Sheets dashboard ensures that these insights remain **live and actionable**, not just a one-time report.

17. Appendix

- **Data Dictionary:** table of all 16 columns with type, description, and sample values (from `Data-Dictionary.csv`).
- **Extra Charts:** sector–state heatmaps, applications distribution, company size breakdowns beyond main report.
- **Transformation Logic:** key Google Sheets formulas (REGEXEXTRACT, PROPER, SPLIT, value conversions).
- **Contribution Matrix:** team-member wise contributions to sourcing, cleaning, KPIs, dashboard, report, and PPT.